

The Matter Mystery

A Story About Matter



Nia and Caleb sat at the science table with Ms. Romero. They had an ice cube, a glass of water, and a balloon.

Nia tapped the ice. "It's hard!"

Caleb splashed the water. "It can move!"

Nia held the balloon. "It's filled with something I can't see or hold!"

Ms. Romero smiled. "Great observations! These are all examples of matter."



Matter Fact

Matter is anything that has mass and takes up space.

Everything Around Us Is Matter

A Story About Matter



Nia looked around the room. “Is the desk matter?” she asked. “Mm-hmm,” said Ms. Romero. “So are your pencils, your books, and the water in your bottle.”

Caleb squeezed the balloon. “What about the air inside this?”

Ms. Romero nodded. “That is matter too. You may not see it easily, but it still takes up space.”

Soon Nia and Caleb realized that matter is almost everywhere around them.



Matter Fact

Matter is anything that has mass and takes up space.

Solids Keep Their Shape

A Story About Matter



Ms. Romero placed an ice cube, a wooden block, a book, and an apple on the table.

"These are all solids," she said.

Nia tapped the block. "It stays the same shape no matter where I put it."

"Exactly," said Ms. Romero. "A solid keeps its own shape. It does not spread out or pour."

Caleb grinned. "So solids are the shape-keepers of matter!"



Matter Fact

A solid keeps its shape because its particles stay packed closely together.

Liquids Flow and Fit

A Story About Matter



Next, Ms. Romero poured water into a tall glass. Then she poured it into a short cup.

“Look!” said Caleb. “The water changed shape!”

“Right,” said Ms. Romero. “A liquid does not keep its own shape. It flows and takes the shape of its container.”

Nia watched the water move. “So a liquid can change shape, but it still stays the same kind of matter.”

Ms. Romero smiled. “You are thinking like scientists.”



Matter Fact

A liquid flows and takes the shape of its container, but it keeps the same amount of matter.

Gases Spread Out

A Story About Matter



Caleb held up the red balloon again. “The air inside must be a gas,” he said.

Ms. Romero nodded. “Yes. A gas spreads out to fill the space around it.”

Nia breathed onto a small mirror and watched it fog up. “So gases can move all around us even when we cannot see them well.”

“That is right,” said Ms. Romero. “Gases do not have a fixed shape or a fixed amount of space.”

The room suddenly felt full of invisible science.



Matter Fact

A gas spreads out to fill the space around it and does not have a fixed shape.

Science Element

A Story About Matter



By the end of the lesson, Nia and Caleb had solved the matter mystery.

Solids keep their shape. Liquids flow and fit their containers. Gases spread out and fill the space around them.

They also learned that matter can change from one state to another. Ice can melt into liquid water. Liquid water can evaporate into gas. Gas can cool and condense back into liquid.

Nia smiled. “Matter can change form, but it is still matter.”

Caleb nodded. “Science is everywhere—even in an ice cube, a glass of water, and a balloon.”



Science Element

Matter is anything that has mass and takes up space. The three main states of matter are solid, liquid, and gas, and matter can change from one state to another.